SITAGLIPTIN

Weight-neutral

Renal dose adjustment

CHEMICAL NAME (R)-4-oxo-4-[3-(trifluoromethyl)-5,6-dihydro[1,2,4]triazolo[4,3-a]pyrazin-7(8H)-yl]-1-(2,4,5-trifluorophenyl)butan-2-amine (phosphate)

CLASS Dipeptidyl peptidase-4 (DPP-4) inhibitor (gliptin)

TRADE NAMES Januvia · Janumet (with metformin) · Xelevia · Tesavel

THERAPEUTIC USES / INDICATIONS

- Type 2 diabetes — add-on to metformin, sulfonylurea, insulin; or monotherapy when metformin unsuitable.
- Useful in elderly (low hypoglycaemia risk).

ADMINISTRATION

- **PO** once daily with or without food.

DOSAGES

- **eGFR ≥45:** 100 mg once daily.
- **eGFR 30–44:** 50 mg once daily.
- **eGFR <30 / dialysis:** 25 mg once daily.

PHARMACOKINETICS

- **Onset:** 1 h. **Duration:** 24 h.
- Minimal metabolism.
- Excreted ~80% unchanged in urine. **t½:** 12 h.

NURSING MANAGEMENT

- Check renal function for dose.
- Monitor HbA1c.
- Report persistent severe abdominal pain (pancreatitis).
- Inspect skin for blisters.

PHARMACODYNAMICS

- Inhibits **DPP-4** → ↑ endogenous incretins (GLP-1, GIP) → glucose-dependent ↑ insulin, ↓ glucagon.
- Low hypoglycaemia risk; weight-neutral.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Headache, nasopharyngitis.
- Nausea.
- Arthralgia.

ADVERSE EFFECTS (SERIOUS)

- Acute pancreatitis.
- Bullous pemphigoid.
- Severe joint pain.
- Hypoglycaemia with sulfonylurea/insulin; hypersensitivity (angioedema, SJS).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity; type 1 diabetes, DKA.
- **Caution:** history of pancreatitis.

KEY INTERACTIONS

- **Sulfonylureas, insulin:** reduce their dose to avoid hypoglycaemia.
- **Digoxin:** slight ↑ levels.
- **ACE inhibitors:** ↑ angioedema risk.

PATIENT EDUCATION

- Take once daily.
- Report severe stomach pain radiating to back, blisters on skin.
- Low risk of hypos unless combined with sulfonylurea or insulin.

KEY POINTS

- DPP-4 inhibitor; boosts incretins.
- Weight-neutral; low hypo risk.
- 100 mg OD; reduce in renal impairment.
- Pancreatitis, pemphigoid.

SOURCES CHECKED **[610]** Sitagliptin 100 mg film coated tablets (UK SmPC) **[611]** Sitagliptin, oral (US label) **[612]** Sitagliptin, oral (US label, Zydus Lifesciences Limited)

Full references p. 432–457